Gabriel Solano

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EDUCATION

University of Pittsburgh, Swanson School of Engineering Expected May 2029

B.S. Electrical Engineering | Minors: Industrial Engineering and Computer Engineering

Certificate (in progress): Innovation, Product Design, and Entrepreneurship

Awards: The Gates Scholarship: 1 of 750 recipients selected from 50,000+ applicants nationally to receive a last dollar scholarship from The Gates Foundation

SKILLS

Languages: C++, Java, MATLAB, VHDL, C

Tools: Quartus Prime, ModelSim, Excel, Fusion 360

Hardware: Intel MAX 10 FPGA, 74HC logic ICs, Arduino, SPI, accelerometer interfacing, breadboard prototyping

Lab: Oscilloscopes, waveform generators, multimeters; RC, RL, RLC, and AC circuit measurement

Certification: Autodesk Fusion Certified User, May 2025

TECHNICAL PROJECTS

Digital Systems in VHDL, *Digital Circuits and Systems* Oct 2026

- Designed a hierarchical binary-to-decimal and hexadecimal converter in VHDL that drives three seven-segment displays on an Intel MAX 10 FPGA, deriving the display decoder logic with K-maps
- Built an 8-bit ripple-carry adder from full adders, validating hand-calculated critical-path delays (50 ns carry-out, 51 ns sum) in ModelSim, and a BCD adder that shows two-digit sums on seven-segment displays
- Implemented a rock-paper-scissors judge and 4:1 multiplexers in dataflow and behavioral VHDL; verified with Tcl-scripted stimulus in ModelSim covering all 16 judge input combinations

Auto-EQ, *High School Capstone* Oct 2024 - May 2025

- Designed an ambient-level-adaptive gain controller using an SPI digital potentiometer as a voltage divider with 256 attenuation steps
- Implemented exponential moving-average filtering in embedded C++ to smooth gain changes; validated against simulated sound input, as hardware constraints limited live microphone integration

Concussion Sensing Device, *Career Technical Engineering Course Project* Dec 2022 - May 2023

- Built an Arduino and accelerometer head-impact detector with a two-tier LED severity output based on g-force thresholds
- Tested with controlled drop heights; the LED tier matched the expected severity in 90 of 100 trials

Discrete Logic Design, *Digital Circuits and Systems* Sep 2026

- Designed and built a 2-bit by 2-bit binary multiplier on breadboards from 74HC-series CMOS ICs, minimizing logic with K-maps; verified all 16 input combinations on LEDs against the truth table
- Designed and built a 2-bit rock-paper-scissors judge from four 74HC-series ICs with push-button inputs and LED outputs, handling ties and invalid selections; later reimplemented in VHDL on an FPGA

EXPERIENCE

Career Technical Engineering, *Shadow and Student* Aug 2022 - Jun 2025

- Trained under professional engineers on mechanical, electrical, and aerospace projects, using laser cutters, mills, drill presses, and saws
- Gained experience in circuitry, machining, and product development, producing the Concussion Sensing Device and Auto-EQ

Colossal Construction, *Renovation Team Member* Jun 2023 - Aug 2026

- Enhanced turnaround time of projects by 20%
- Assisted in 5 residential renovation projects, operating power tools safely and effectively, increasing on-site productivity by 30%

LEADERSHIP

National Society of Black Engineers, *Conference Planning Chair* Sep 2025 - Present

- Collaborated with the Financial Zone to streamline operational workflows, improving member-data collection and fundraising efficiency for a group of 75+ members
- Contributed to fundraising efforts that generated $8,000+ in sales through coordinated outreach and event support
- Organized conference for 50+ members to ensure chapter functionality and increase retention

Studio-605, *Production Team Internship Project* Oct 2023 - May 2024

- Led a team of 5 to build a multifunctional film and audio studio space for up to 10 students